Inverted Pendulum: Modelling, Linearisation and PID Control

ELE2038 Signals and Control - Control Lab Assignment

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Abstract. In this report the inverted pendulum on a cart is linearised about its upright equilibrium, producing transfer functions $G_\theta(s)$ and $G_x(s)$, both shown to be BIBO unstable with a right half-plane pole at $s \approx 5.09\,\mathrm{rad/s}$. All open-loop responses (impulse, step, sinusoidal, and decaying sinusoidal) diverge, confirming instability is a property of the system rather than the input. A PID controller with $K_p = -400$, $K_d = -50$, $K_i = -100$ is introduced to keep the pendulum upright within a specified angle on the linearised model; Routh's criterion confirms closed-loop BIBO stability and integral action eliminates steady-state offset. Nonlinear simulation reveals sustained oscillations under impulse disturbance and unrecoverable departure from the upright position under step disturbance, demonstrating the controller fails once θ grows large enough that the linearisation $\sin\theta \approx \theta$, $\cos\theta \approx 1$ no longer holds..

1 Linearisation

1.1 System model

The system is an inverted pendulum on a cart of mass M, with rod mass m and half-length ℓ. State variables: $x_1 = x$, $x_2 = \dot{x}$, $x_3 = \theta$, $x_4 = \dot{\theta}$. The nonlinear equations of motion $\dot{x}_2 = \phi$, $\dot{x}_4 = \psi$ are:

$$\phi = \frac{4m\ell x_4^2 \sin x_3 + 4F - 3mg \sin x_3 \cos x_3}{4(M+m) - 3m\cos^2 x_3}, \tag{1.1}$$

$$\psi = \frac{-3(m\ell x_4^2 \sin x_3 \cos x_3 + F\cos x_3 - (M+m)g\sin x_3)}{[4(M+m) - 3m\cos^2 x_3]\,\ell}. \tag{1.2}$$

as defined in the lab sheet.

1.2 Equilibrium (Q2.1i)

Substituting $(F^e, x_3^e, x_4^e) = (0,0,0)$ into (1.1) (1.2) gives $\phi(0,0,0) = \psi(0,0,0) = 0$, confirming the upright position as an equilibrium. This was verified in the code liked using `sympy`.

1.3 Linearisation (Q2.1ii)

The Laplace transform only applies to linear differential equations because the input-output relationship is linear. It converts $\dot{x} = ax$ into $sX = aX$ The terms $\sin x_3$, $\cos x_3$, and $x_4^2 \sin x_3 \cos x_3$ in (1.1)(1.2) are nonlinear, so no transfer function $X_3(s)/F(s)$ exists for the original system.

Taylor's theorem approximates a smooth function near a point by retaining only the linear (first-order) terms of its expansion, equivalent to replacing the function by its tangent plane at that point. Linearising ϕ and ψ about $(0,0,0)$ and evaluating each partial derivative at the equilibrium (where $\cos 0 = 1$, $\sin 0 = 0$, $x_4 = 0$, reducing the denominator to $\ell(4M+m)$) gives, for ψ:

$$\left.\frac{\partial\psi}{\partial F}\right|_0 = -\frac{3}{\ell(4M+m)}, \tag{1.3}$$

$$\left.\frac{\partial\psi}{\partial x_3}\right|_0 = \frac{3(M+m)g}{\ell(4M+m)}, \tag{1.4}$$

$$\left.\frac{\partial\psi}{\partial x_4}\right|_0 = 0. \tag{1.5}$$

For ϕ:

$$\left.\frac{\partial\phi}{\partial F}\right|_0 = \frac{4}{4M+m}, \tag{1.6}$$

$$\left.\frac{\partial\phi}{\partial x_3}\right|_0 = -\frac{3mg}{4M+m}, \tag{1.7}$$

$$\left.\frac{\partial\phi}{\partial x_4}\right|_0 = 0. \tag{1.8}$$

Defining $c = 3/[\ell(4M+m)]$ and $d = 3(M+m)g/[\ell(4M+m)]$, the linearised dynamics are:

$$\dot{x}_4 = -cF + dx_3, \tag{1.9}$$

$$\dot{x}_2 = \frac{4}{4M+m}F - \frac{3mg}{4M+m}x_3. \tag{1.10}$$

1.4 Transfer functions and BIBO stability (Q2.1iii)

Using the derived linear equations, the Laplace transform of (1.9) with zero initial conditions and $X_4(s) = sX_3(s)$ gives $s^2X_3 = -cF + dX_3$, from which:

$$G_\theta(s) = \frac{X_3(s)}{F(s)} = \frac{-c}{s^2 - d}. \tag{1.11}$$

Since $d > 0$, the pole $s = +\sqrt{d}$ is in the right half-plane, determining that G_θ is not BIBO stable. Physically, any small angular deviation increases the gravitational pull on the rod, increasing the deviation angle rather than restoring the rod to its upright equilibrium. For the cart position transfer function G_x, substituting the expression for X_3 already found from (1.11) into the Laplace transform of (1.10) and simplifying gives:

$$G_x(s) = \frac{4s^2 - 3g/\ell}{(4M+m)\,s^2(s^2 - d)}. \tag{1.12}$$

Double poles at $s = 0$, and another at $s = \pm\sqrt{d}$: confirm that thsi function is also not BIBO stable. The double pole at the origin means the cart drifts unbound under any sustained input.

2 Open-loop responses

2.1 Symbolic responses (Q2.2)

Applying the inverse Laplace transform symbolically to $G_\theta \cdot F(s)$ for each input gives the following. Impulse ($F(s) = 1$):

$$x_3^{\text{imp}}(t) = -\frac{c\sinh(\sqrt{d}\,t)}{\sqrt{d}}\,u(t). \tag{2.1}$$

Step ($F(s) = 1/s$):

$$x_3^{\text{step}}(t) = -\frac{c}{d}[\cosh(\sqrt{d}\,t) - 1]\,u(t). \tag{2.2}$$

Sinusoidal ($F(t) = \sin\omega t$):

$$x_3^{\sin}(t) = \frac{c\left(\sqrt{d}\sin(\omega t) - \omega\sinh(\sqrt{d}\,t)\right)}{\sqrt{d}(d+\omega^2)}\,u(t). \tag{2.3}$$

The first term is bounded; the second grows exponentially. For the decaying sinusoidal input $F(t) = e^{-\lambda t}t\sin\omega t$, the Laplace transform is obtained symbolically as:

$$F(s) = \frac{2\omega(s+\lambda)}{[(s+\lambda)^2 + \omega^2]^2}. \tag{2.4}$$

The response $\mathscr{L}^{-1}\{G_\theta \cdot F(s)\}$ again contains $\sinh(\sqrt{d}\,t)$ terms. Although $F(t)$ tends towards 0 as t tends towards ∞, the output still diverges showing that instability is a property of the system, not the input.

2.2 Numerical responses (Q2.3)

With $M = 0.3$, $m = 0.1$, $\ell = 0.35$, $g = 9.81$:

$$c \approx 6.59, \quad d \approx 25.87, \quad \sqrt{d} \approx 5.09\,\text{rad/s}. \tag{2.5}$$

Figures 1-4 show the step and impulse responses of G_θ and G_x, all diverging as predicted above. The system fails to remain stable with no controller however, the G_x impulse grows more slowly than G_θ during short timeframes due to the double pole at $s = 0$. Figures 5 and 6 show $\theta(t)$ and $x(t)$ respectively under $F(t) = \sin(100t^2)$ over $t \in [0, 0.2]$ s.. Growth begins once the instantaneous frequency $\omega(t) = 200t$ crosses $\sqrt{d} \approx 5.09\,\text{rad/s}$ at $t \approx 0.025\,\text{s}$.

Figure 1: Step response of G_θ.

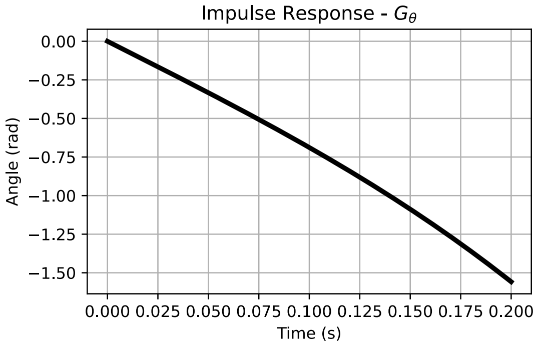

Figure 2: Impulse response of G_θ.

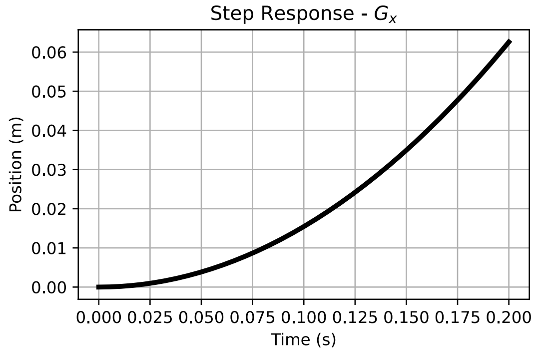

Figure 3: Step response of G_x.

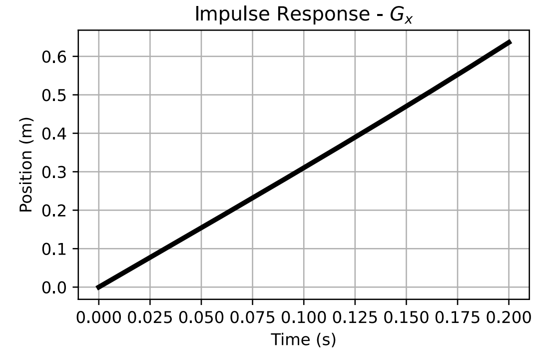

Figure 4: Impulse response of G_x. Linear growth at short times is consistent with the double pole at $s = 0$, which contributes a $t \cdot u(t)$ term alongside the sinh contribution.

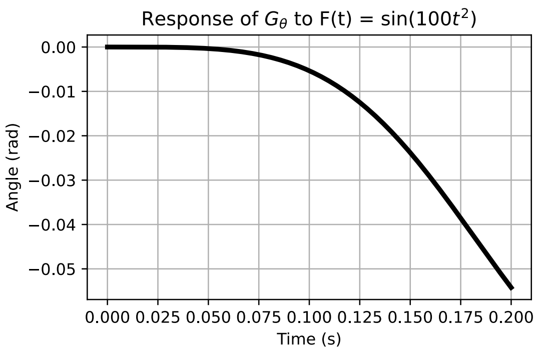

Figure 5: $\theta(t)$ under $F(t) = \sin(100t^2)$, $t \in [0, 0.2]$ s.

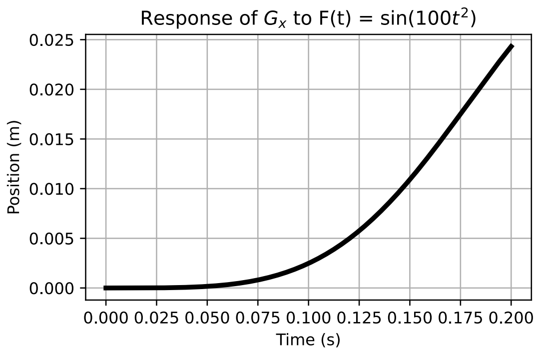

Figure 6: $x(t)$ under $F(t) = \sin(100t^2)$, $t \in [0, 0.2]$ s. Growth begins once the instantaneous frequency crosses $\sqrt{d} \approx 5.09$ rad/s at $t \approx 0.025$ s.

3 PID controller design

3.1 Closed-loop structure and characteristic polynomial

The closed-loop block diagram is shown in Figure 7. With PID controller $G_c(s) = K_p + K_i/s + K_d s$ and disturbance D at the plant input, the load transfer function is $G_{\text{load}} = G_\theta/(1 + G_c G_\theta)$. Substituting (1.11) and multiplying through by $s(s^2 - d)/(-c)$:

$$p(s) = s^3 - cK_d s^2 - (d + cK_p)s - cK_i. \qquad (3.1)$$

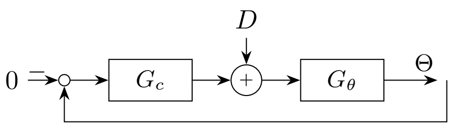

Figure 7: Closed-loop block diagram. $D(s)$ at plant input; $\Theta^{\text{sp}} = 0$.

3.2 Gain selection and impulse response (Q2.4i)

Gains $K_p = -400$, $K_d = -50$, $K_i = -100$ were tuned to satisfy $|\theta| \leq 5°$ and $|\theta| \leq 0.1°$ for $t \geq 0.4$ s following $D(s) = 1$. The maximum $|\theta|$ after 0.4 s was confirmed numerically as 0.0309°, as can be seen in the Python simulation linked. Figures 8 and 9 show $\theta(t)$ and $F(t)$ respectively.

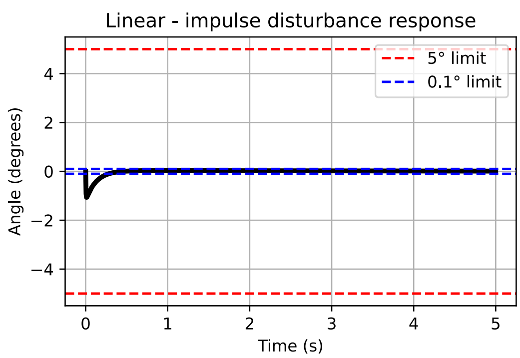

Figure 8: Angle $\theta(t)$ under $D(s) = 1$. Specification met: $|\theta| \leq 0.1°$ for $t \geq 0.4$ s.

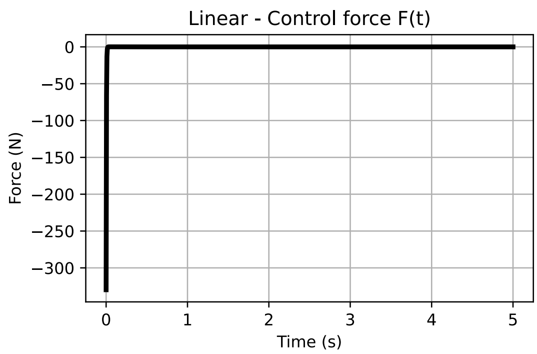

Figure 9: Control force $F(t)$ under $D(s) = 1$.

3.3 Routh's stability analysis (Q2.4iv)

With $c \approx 6.59$, $d \approx 25.87$ and the selected gains:

$$p(s) = s^3 + 329.5s^2 + 2610.1s + 659. \qquad (3.2)$$

Routh's tabulation:

$$\begin{array}{c|cc} s^3 & 1 & 2610.1 \\ s^2 & 329.5 & 659 \\ s^1 & b_1 & 0 \\ s^0 & 659 & 0 \end{array} \qquad (3.3)$$

where $b_1 = (329.5 \times 2610.1 - 659)/329.5 \approx 2608.1$. All first-column entries $[1,\ 329.5,\ 2608.1,\ 659]$ are positive: the closed-loop system is BIBO stable.

3.4 Zero steady-state offset (Q2.4ii)

By the final value theorem, the steady-state angle under a unit step disturbance is $\lim_{t\to\infty} \theta(t) = \lim_{s\to 0^+} G_{\text{load}}(s)$. Substituting the equation:

$$\lim_{s\to 0^+} \frac{-cs}{s^3 + 329.5s^2 + 2610.1s + 659} = \frac{0}{659} = 0. \qquad (3.4)$$

The integral term in the PID controller ensures the steady-state angle error is exactly zero, regardless of the specific value of K_i. This is confirmed by Figure 10.

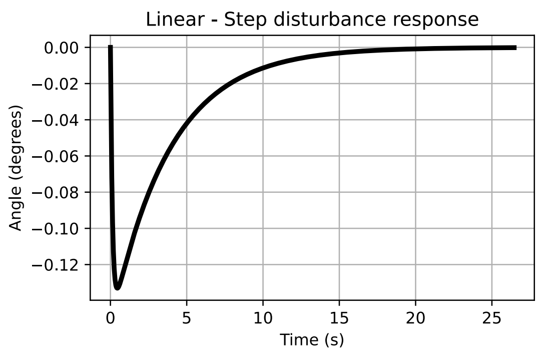

Figure 10: Step disturbance response. $\theta \to 0$, confirming zero offset.

3.5 Parameter sensitivity (Q2.4iii)

Figures 11-13 show the closed-loop impulse response under $\pm 100\%$ misestimation of M, m, and ℓ. All responses remain stable. The large gains provide robustness: parameter variation shifts c and d but not enough to produce a sign change in the Routh first column.

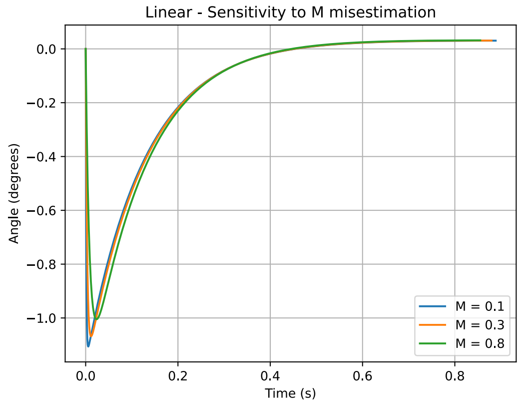

Figure 11: Sensitivity to M misestimation ($\pm 100\%$).

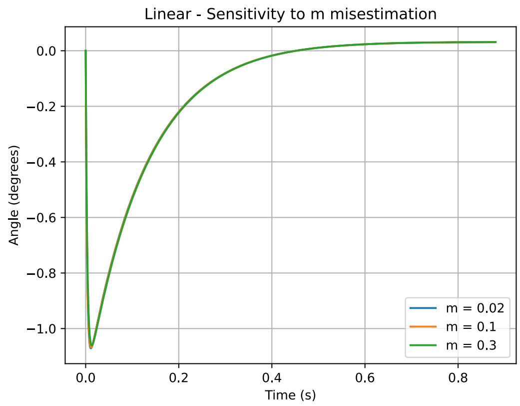

Figure 12: Sensitivity to m misestimation ($\pm 100\%$).

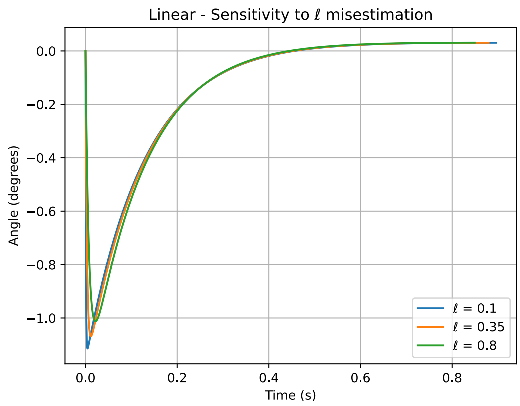

Figure 13: Sensitivity to ℓ misestimation ($\pm 100\%$). All responses remain stable, showing little variation between values plotted, demonstrating sufficient system robustness in a practical scenario against modelling error.

4 Advanced analysis

4.1 Cart position under angle control (Q2.5i)

Because the PID controller leaves the double pole at $s = 0$ in G_x unaddressed, the cart is free to drift without bound, as shown in Figure 14. Controlling both variables simultaneously would require a multi-loop strategy, where an outer loop generates a reference angle θ_{sp} from the cart position error, which the inner PID loop then tracks.

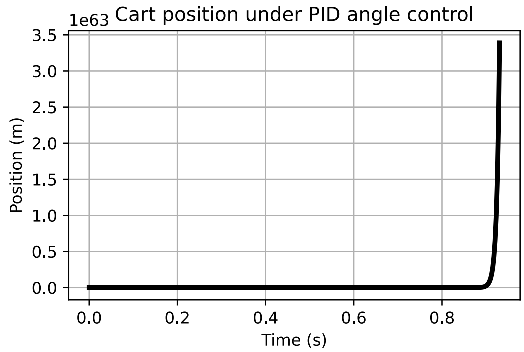

Figure 14: Cart position under PID control.

4.2 Nonlinear simulation (Q2.5ii)

The nonlinear system is formulated as an IVP, with an integral term introduced as a fifth state variable $z = \int_0^t x_3\, d\tau$, $\dot{z} = x_3$ (see linked code). The impulse is approximated as a rectangular force pulse of 1000 N over the first millisecond, giving unit impulse area and preserving $D(s) \approx 1$.

Figure 15 shows that the PID controller, designed for the linearised model, fails on the nonlinear system. Under impulse disturbance the pendulum oscillates rather than settling to zero. Under step disturbance the pendulum departs the upright position unrecoverably. Sensitivity plots confirm this behaviour across all parameter variations. Once θ grows large enough that $\sin\theta \approx \theta$ and $\cos\theta \approx 1$ break down, the nonlinear terms dominate and the controller can't recover.

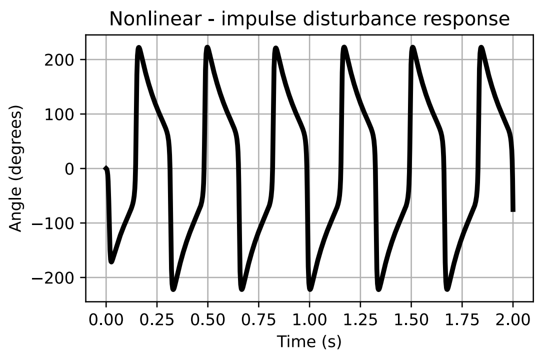

Figure 15: Nonlinear angle $\theta(t)$ under impulse disturbance.

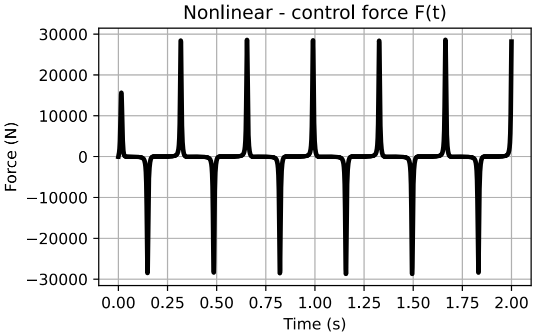

Figure 16: Nonlinear control force $F(t)$ under impulse disturbance. The oscillation confirms unsuccessful damping of motion.

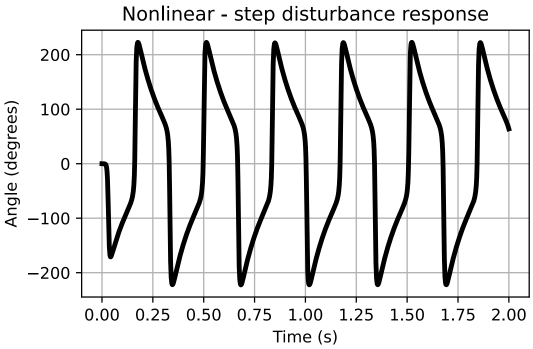

Figure 17: Nonlinear response to step disturbance.

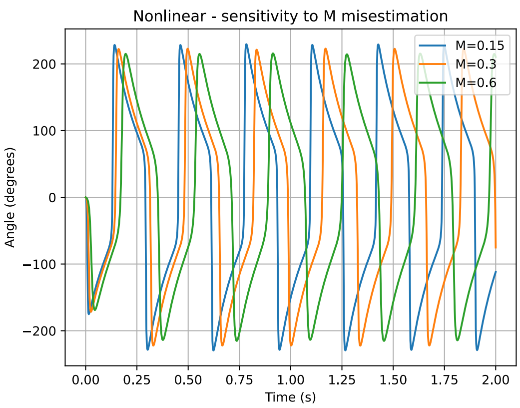

Figure 18: Nonlinear sensitivity to M misestimation.

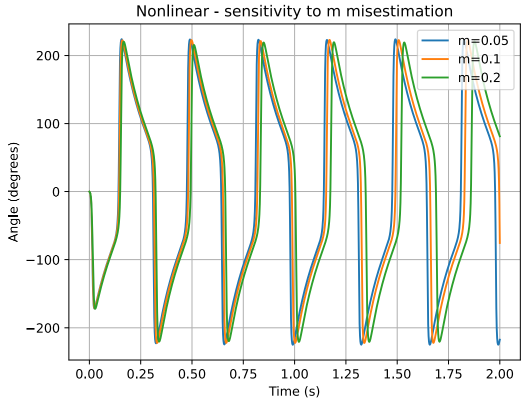

Figure 19: Nonlinear sensitivity to m misestimation.

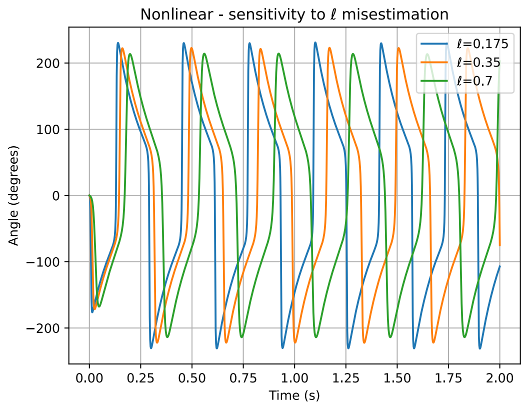

Figure 20: Nonlinear sensitivity to ℓ misestimation. All three cases show the same oscillatory failure mode regardless of parameter values.

Conclusions

The inverted pendulum was linearised about its upright equilibrium, providing transfer functions G_θ and G_x that are BIBO unstable due to a right half-plane pole at $s \approx 5.09$ rad/s. All open-loop responses diverge regardless of input type.

A PID controller was designed to stabilise the system. Routh's criterion confirms closed-loop stability, the integral term guarantees zero steady-state offset, and the controller proves robust to significant misestimation of M, m, and ℓ.

Nonlinear simulation reveals the limits of this design. Once $\sin\theta \approx \theta$ and $\cos\theta \approx 1$ no longer hold, the PID loses control, causing unrecoverable departure from the upright position under step disturbance.

Additionally, because the controller was designed around G_θ alone, the cart drifts without bound. Resolving this would require a multi-loop strategy, with an outer loop regulating cart position by generating a reference angle for the inner PID loop to track.